



Module_1:

Team Members:

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Project Title:

BME 2315, Module 1: Alzheimer's | Educational Attainment as a Factor in Alzheimer's Disease Progression test

Project Goal:

This project seeks to view the correlation between level of education and the development and severity of Alzheimer's disease through the analysis of the Thal score while using Python. Additionally, the project seeks to find if there is a correlation between the ratio of amyloid-beta 40 / amyloid-beta 42 compared to the age of onset symptoms of Alzheimer's Disease.

Disease Background:

Fill in information about 11 bullets:

- Prevalence & incidence: In 2017, approximately 6.08 million Americans either had clinical Alzheimer's disease (AD) or mild cognitive impairment resulting from AD (<https://www.sciencedirect.com/science/article/abs/pii/S155252601733813X>) According to the Alzheimer's Association, over 7 million Americans are currently living with Alzheimer's, and this number is projected to grow to 13 million in the next 25 years (<https://www.alz.org/alzheimers-dementia/facts-figures>).
- Economic burden: The total annual cost (per capita) of AD ranged from \$468.28 (in mild cases of AD) to \$171,283.80 (in severe cases of AD) (prices in U.S. Dollars). Factors & Facts: Cost of care raised non-linearly with disease severity. Indirect caregiver cost was the main contributor to societal cost (<https://www.sciencedirect.com/science/article/pii/S2212109923000948>)
- Risk factors (genetic, lifestyle): Genetic Risk Factors: APOE gene alleles:

APOE ϵ 4 → increases AD risk, associated with earlier onset; APOE ϵ 3 → neutral effect; APOE ϵ 2 → reduces AD risk, delays onset. Early-Onset AD (EOAD) genes: APP (Amyloid precursor protein); PSEN1 (Presenilin 1); PSEN2 (Presenilin 2). Other genetic associations: Elevated LDL cholesterol levels linked to EOAD; Genome-wide studies identifying 75+ risk loci and new genes related to inflammation, lipid metabolism, and neuronal function. Lifestyle/Environmental Risk Factors: Negative (increase AD risk): Western diet (high in meat, ultraprocessed foods, saturated/trans fats, refined carbs); Obesity (especially midlife); Hypertension, high cholesterol, and diabetes mellitus; High alcohol intake and smoking; Low levels of B vitamins (B6, B12, folate), leading to elevated homocysteine; Physical inactivity; Social isolation and depression; Hearing loss (midlife); Air pollution; Traumatic brain injury (TBI); Occupational exposure to heavy metals; Certain infections (e.g., herpes simplex virus). Positive (reduce AD risk): Mediterranean diet (MeDi) → high in fruits, vegetables, legumes, whole grains, fish, and olive oil; low in meat; Vegetarian/vegan diet (similar benefits); Specific foods/supplements: Coffee, fruits/vegetables, legumes, whole grains, fish, omega-3 fatty acids, antioxidant vitamins (A, C, E), vitamin D; Physical activity (regular exercise); Cognitive stimulation & education (lifelong learning, complex work, mental activity); Social engagement & strong networks; Moderate alcohol intake (mainly wine, in the context of MeDi); Hormone replacement therapy (HRT) (beneficial in some cases); Statins, antihypertensives, NSAIDs (potential protective roles). (generated using chatGPT)

- Societal determinants: There are limited societal determinants that link to Alzheimer's; however, research shows that individuals that have lower education levels are more prone to Alzheimer's disease than those with higher education levels. In addition, those who have greater access to health care are less prone to worsening Alzheimer's due to greater access to quality care. (<https://www.cdc.gov/alzheimers-dementia/php/sdoh/index.html>)
- Symptoms: The biggest symptom of Alzheimer's disease is memory loss, which gets worse and worse as the disease progresses. At the beginning, this memory loss could be as simple as general cluelessness, but can get to a point of forgetting identity. This memory loss often leads to issues involving making decisions, reasoning clearly, and personality changes. (<https://www.mayoclinic.org/diseases->

[conditions/alzheimers-disease/symptoms-causes/syc-20350447](https://www.mayoclinic.org/diseases-conditions/alzheimers-disease/symptoms-causes/syc-20350447))

- Diagnosis: Neurologists usually diagnose Alzheimer's; however, geriatricians or other healthcare professionals can help with diagnosis. Family members and close friends of the patient are interviewed to provide ideas about the nature of the patient's potential memory loss. The healthcare professional looks for loss of cognitive skills, but more importantly, the impact it has on the patient's everyday life: severity, ability to function, and personality changes. Diagnosing Alzheimer's: How Alzheimer's is diagnosed - Mayo Clinic: <https://www.mayoclinic.org/diseases-conditions/alzheimers-disease/in-depth/alzheimers/art-20048075>
- Standard of care treatments (& reimbursement): There are no cures for the disease, so the biggest ways the disease is treated is through medicine and therapies. Common medicines used to treat Alzheimer's include Memantine (which helps with memory), SSRIs (to reduce irritability), and some select antipsychotics. In terms of therapies, those with Alzheimer's can go through reminiscence therapies (to try and bring back old memories), cognitive stimulation therapies, and reality orientation training. Reimbursement for this care is provided through Medicare and other insurance plans. (https://www.alz.org/getmedia/953daa9b-3a95-4c9c-8225-314fcb700d62/cognitive-impairment-care-planning-toolkit_1.pdf, <https://www.webmd.com/alzheimers/dementia-treatments-overview>)
- Disease progression & prognosis: Alzheimer's can eventually lead to death through other conditions, but the timeline from diagnosis to death can vary by around 20 years. The first stage of Alzheimer's is preclinical Alzheimer's, which can only be diagnosed in research settings, as symptoms are not visible to either the patient or those surrounding them. Following this is mild cognitive impairment [MCI] due to Alzheimer's, which is characterized by forgetting small items, but the patient can remember again relatively easily through conversations. In the middle of the scale is mild dementia, which is where memory loss, judgment issues, personality changes, expression. Alzheimer's stages: How the disease progresses - Mayo Clinic: <https://www.mayoclinic.org/diseases-conditions/alzheimers-disease/in-depth/alzheimers-stages/art-20048448>
- Continuum of care providers: The treatment of Alzheimer's is very

patient-centered, meaning that a wide variety of care providers are useful for the patient and/or their family. Aside from physicians and nurses, other care providers involved include psychologists, social workers, and direct care workers. All of these healthcare professionals collaborate to provide personal care, administer therapies, and make the patient feel comfortable. (<https://www.alz.org/professionals/professional-providers>)

- Biological mechanisms (anatomy, organ physiology, cell & molecular physiology): Generally, Alzheimer's Disease initially affects the brain neurons, where it destroys connections in memory-related areas of the brain (such as the hippocampus and the entorhinal cortex). As the disease progresses, the cerebral cortex is attacked (impacting motor skills, language, and thinking). The disease continues to progress until it eventually becomes fatal. <https://www.nia.nih.gov/health/alzheimers-causes-and-risk-factors/what-happens-brain-alzheimers-disease>
- Clinical Trials/next-gen therapies: The National Institute on Aging (NIA) has 495 ongoing clinical trials to find better treatments and possibly a cure for Alzheimer's Disease. Among these are 68 pharmacological trials, 157 non-pharmacological trials, 217 dementia care and caregiving trials, 31 trials to understand disease development and progress, 19 studies on diagnostic tools and assessment, and 3 trials on neuropsychiatric symptoms. By casting a broad net of research, these trials can lead to a wide variety of potential treatments for the disease, creating a more holistic treatment plan. [https://www.nia.nih.gov/research/ongoing-AD-trials#:~:text=The%20National%20Institute%20on%20Aging%20%28NIA%](https://www.nia.nih.gov/research/ongoing-AD-trials#:~:text=The%20National%20Institute%20on%20Aging%20%28NIA%20)

Data-Set:

This dataset comes from the Seattle Alzheimer's Disease Brain Cell Atlas (SEA-AD), described in "Integrated multimodal cell atlas of Alzheimer's disease" (Nature Neuroscience, 2024). This dataset includes information of postmortem brain tissue from 84 donors. Donors were of both sexes, and had an average age of 88). The total compilation of tissue spanned the entire Alzheimer pathology spectrum. Data was collected using methods such as quantitative neuropathology, single-nucleus RNA sequencing (snRNA-seq), ATAC-seq, spatial transcriptomics, and multiome profiling. Measurements included counts of plaques/tangles per area, transcript count per nucleus, accessibility peaks of chromatin, and cognitive exam scores. All

data is publicly accessible through the SEA-AD portal. Source: Integrated multimodal cell atlas of Alzheimer's disease (Hodge et al., 2024)

Data Analysis:

Below is Python code used for initial analysis/organization. More detailed analysis in progress.

The SEA-AD dataset was used to examine relationships between the patients' level of education and the progression of AD. Python was used to load and combine metadata and biomarker measurements, sort patients based on their education levels, and analyze Thal scores across the entire group. Bar charts were used to assist with the understanding of patterns within the dataset, and an ANOVA statistical test was used to determine if differences in Thal scores between the various education levels were significant.

(Describe how you analyzed the data. This is where you should intersperse your Python code so that anyone reading this can run your code to perform the analysis that you did, generate your figures, etc.)

```
In [30]: %pip install matplotlib
          %pip install scipy
          %pip install pandas
          %pip install scikit-learn
```

```
Requirement already satisfied: matplotlib in c:\users\rmslo\appdata\local\programs\python\python313\lib\site-packages (3.10.6)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\rmslo\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\users\rmslo\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\rmslo\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (4.60.0)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\rmslo\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (1.4.9)
Requirement already satisfied: numpy>=1.23 in c:\users\rmslo\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (2.3.3)
Requirement already satisfied: packaging>=20.0 in c:\users\rmslo\appdata\roaming\python\python313\site-packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in c:\users\rmslo\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\rmslo\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\rmslo\appdata\roaming\python\python313\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in c:\users\rmslo\appdata\roaming\python\python313\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

```

[notice] A new release of pip is available: 24.3.1 -> 25.2
[notice] To update, run: python.exe -m pip install --upgrade pip
Requirement already satisfied: scipy in c:\users\rmslo\appdata\local\programs\python\python313\lib\site-packages (1.16.2)
Requirement already satisfied: numpy<2.6,>=1.25.2 in c:\users\rmslo\appdata\local\programs\python\python313\lib\site-packages (from scipy) (2.3.3)
Note: you may need to restart the kernel to use updated packages.

[notice] A new release of pip is available: 24.3.1 -> 25.2
[notice] To update, run: python.exe -m pip install --upgrade pip
Requirement already satisfied: pandas in c:\users\rmslo\appdata\local\programs\python\python313\lib\site-packages (2.3.3)
Requirement already satisfied: numpy>=1.26.0 in c:\users\rmslo\appdata\local\programs\python\python313\lib\site-packages (from pandas) (2.3.3)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\rmslo\appdata\local\programs\python\python313\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\rmslo\appdata\local\programs\python\python313\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\rmslo\appdata\local\programs\python\python313\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: six>=1.5 in c:\users\rmslo\appdata\roaming\python\python313\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.

[notice] A new release of pip is available: 24.3.1 -> 25.2
[notice] To update, run: python.exe -m pip install --upgrade pip
Requirement already satisfied: scikit-learn in c:\users\rmslo\appdata\local\programs\python\python313\lib\site-packages (1.7.2)
Requirement already satisfied: numpy>=1.22.0 in c:\users\rmslo\appdata\local\programs\python\python313\lib\site-packages (from scikit-learn) (2.3.3)
Requirement already satisfied: scipy>=1.8.0 in c:\users\rmslo\appdata\local\programs\python\python313\lib\site-packages (from scikit-learn) (1.16.2)
Requirement already satisfied: joblib>=1.2.0 in c:\users\rmslo\appdata\local\programs\python\python313\lib\site-packages (from scikit-learn) (1.5.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in c:\users\rmslo\appdata\local\programs\python\python313\lib\site-packages (from scikit-learn) (3.6.0)
Note: you may need to restart the kernel to use updated packages.

[notice] A new release of pip is available: 24.3.1 -> 25.2
[notice] To update, run: python.exe -m pip install --upgrade pip

```

In [31]: **import** csv

```

with open("UpdatedMetaData.csv", newline="") as f:
    reader = csv.reader(f)
    headers = next(reader)
    for h in headers:
        print(h)

with open("UpdatedLuminex.csv", newline="") as f:
    reader = csv.reader(f)
    headers = next(reader)
    for h in headers:
        print(h)

```

Donor ID
Primary Study Name
Secondary Study Name
Age at Death
Sex
Race (choice=White)
Race (choice=Black/ African American)
Race (choice=Asian)
Race (choice=American Indian/ Alaska Native)
Race (choice=Native Hawaiian or Pacific Islander)
Race (choice=Unknown or unreported)
Race (choice=Other)
specify other race
Hispanic/Latino
Highest level of education
Years of education
APOE Genotype
Cognitive Status
Age of onset cognitive symptoms
Age of Dementia diagnosis
Known head injury
Have they had neuroimaging
Consensus Clinical Dx (choice=Alzheimers disease)
Consensus Clinical Dx (choice=Alzheimers Possible/ Probable)
Consensus Clinical Dx (choice=Ataxia)
Consensus Clinical Dx (choice=Corticobasal Degeneration)
Consensus Clinical Dx (choice=Control)
Consensus Clinical Dx (choice=Dementia with Lewy Bodies/ Lewy Body Disease)
Consensus Clinical Dx (choice=Frontotemporal lobar degeneration)
Consensus Clinical Dx (choice=Huntingtons disease)
Consensus Clinical Dx (choice=Motor Neuron disease)
Consensus Clinical Dx (choice=Multiple System Atrophy)
Consensus Clinical Dx (choice=Parkinsons disease)
Consensus Clinical Dx (choice=Parkinsons Cognitive Impairment - no dementia)
Consensus Clinical Dx (choice=Parkinsons Disease Dementia)
Consensus Clinical Dx (choice=Prion)
Consensus Clinical Dx (choice=Progressive Supranuclear Palsy)
Consensus Clinical Dx (choice=Taupathy)
Consensus Clinical Dx (choice=Vascular Dementia)
Consensus Clinical Dx (choice=Unknown)
Consensus Clinical Dx (choice=Other)
If other Consensus dx, describe
Last CASI Score
Interval from last CASI in months
Last MMSE Score
Interval from last MMSE in months
Last MOCA Score
Interval from last MOCA in months
PMI
Rapid Frozen Tissue Type
Ex Vivo Imaging
Fresh Brain Weight
Brain pH
Overall AD neuropathological Change

Thal
Braak
CERAD score
Overall CAA Score
Highest Lewy Body Disease
Total Microinfarcts (not observed grossly)
Total microinfarcts in screening sections
Atherosclerosis
Arteriolosclerosis
LATE
RIN
Severely Affected Donor
Donor ID
ABeta40 pg/ug
ABeta42 pg/ug
tTAU pg/ug
pTAU pg/ug

In [32]: *#1) BUILD OUR CLASS BY COMBINING TWO .csv FILES OF DATA*

```
import csv
import warnings
import matplotlib.pyplot as plt

class Patient:

    all_patients = []
    education_lvl = {}

    def __init__(self, DonorID, ABeta40: float , ABeta42: float, tTau: float,

        self.DonorID = DonorID
        self.ABeta40 = ABeta40
        self.ABeta42 = ABeta42
        self.tTau = tTau
        self.pTau = pTau
        self.sex = None
        self.death_age = None
        self.ed_lvl = None
        self.cog_stat = None
        self.age_symp_on = None
        self.age_diag = None
        self.head_inj = None
        self.thal_score = None
        Patient.all_patients.append(self)

    def __repr__(self):
        return f"{self.DonorID} | sex: {self.sex} | ABeta40 {self.ABeta40} | t

    def get_id(self):
        return self.DonorID

    def get_ABeta42(self):
        return self.ABeta42
```

```

def get_thal(self):
    return self.thal_score

def get_death_age(self):
    return self.death_age

@classmethod
def combine_data(cls, filename: str):
    with open(filename, encoding="utf8") as f:
        reader = csv.DictReader(f)
        rows_of_patients = list(reader)
        for row in range(len(rows_of_patients)):
            if Patient.all_patients[row].DonorID == rows_of_patients[row][
                "Donor ID"]:
                if rows_of_patients[row]["Sex"] != "":
                    Patient.all_patients[row].sex = rows_of_patients[row][
                        "Sex"]

                if rows_of_patients[row]["Age at Death"] != "":
                    Patient.all_patients[row].death_age = int(rows_of_patients[
                        row]["Age at Death"])

                if rows_of_patients[row]["Highest level of education"] != "":
                    Patient.all_patients[row].ed_lvl = rows_of_patients[row][
                        "Highest level of education"]

                if rows_of_patients[row]["Cognitive Status"] != "":
                    Patient.all_patients[row].cog_stat = rows_of_patients[row][
                        "Cognitive Status"]

                if rows_of_patients[row]["Age of onset cognitive symptoms"] != "":
                    Patient.all_patients[row].age_symp_on = int(rows_of_patients[
                        row]["Age of onset cognitive symptoms"])

                if rows_of_patients[row]["Age of Dementia diagnosis"] != "":
                    Patient.all_patients[row].age_diag = int(rows_of_patients[
                        row]["Age of Dementia diagnosis"])

                if rows_of_patients[row]["Known head injury"] != "":
                    Patient.all_patients[row].head_inj = rows_of_patients[row][
                        "Known head injury"]

                if rows_of_patients[row]["Thal"] != "":
                    Patient.all_patients[row].thal_score = int(rows_of_patients[
                        row]["Thal"])
            else:
                warnings.warn("IDs do not match.")

@classmethod
def instantiate_from_csv(cls, filename: str, other_file: str):
    with open(filename, encoding="utf8") as f:
        reader = csv.DictReader(f)
        rows_of_patients = list(reader)
        for row in rows_of_patients:
            Patient(
                DonorID = row['Donor ID'],
                ABeta40 = float(row['ABeta40 pg/ug']),
                ABeta42 = float(row['ABeta42 pg/ug']),
                tTau = float(row['tTAU pg/ug']),
                pTau = float(row['pTAU pg/ug'])
            )
        Patient.all_patients.sort(key = Patient.get_id)

```

```

        Patient.combine_data(other_file)

#4) BUILD DICTIONARIES TO DO SORTING AND SUB-SORTING
@classmethod
def sort_ed(cls):
    for patient in Patient.all_patients:
        if patient.ed_lvl is not None:
            Patient.education_lvl.setdefault(patient.ed_lvl, [])
    for patient in Patient.all_patients:
        if patient.ed_lvl is not None:
            Patient.education_lvl[patient.ed_lvl].append(patient)

@classmethod
def subsort_thal(cls):
    for key in Patient.education_lvl:
        values = Patient.education_lvl.get(key)
        values.sort(key = Patient.get_thal)
        Patient.education_lvl.update({key: values})

# ----- MAIN SCRIPT -----

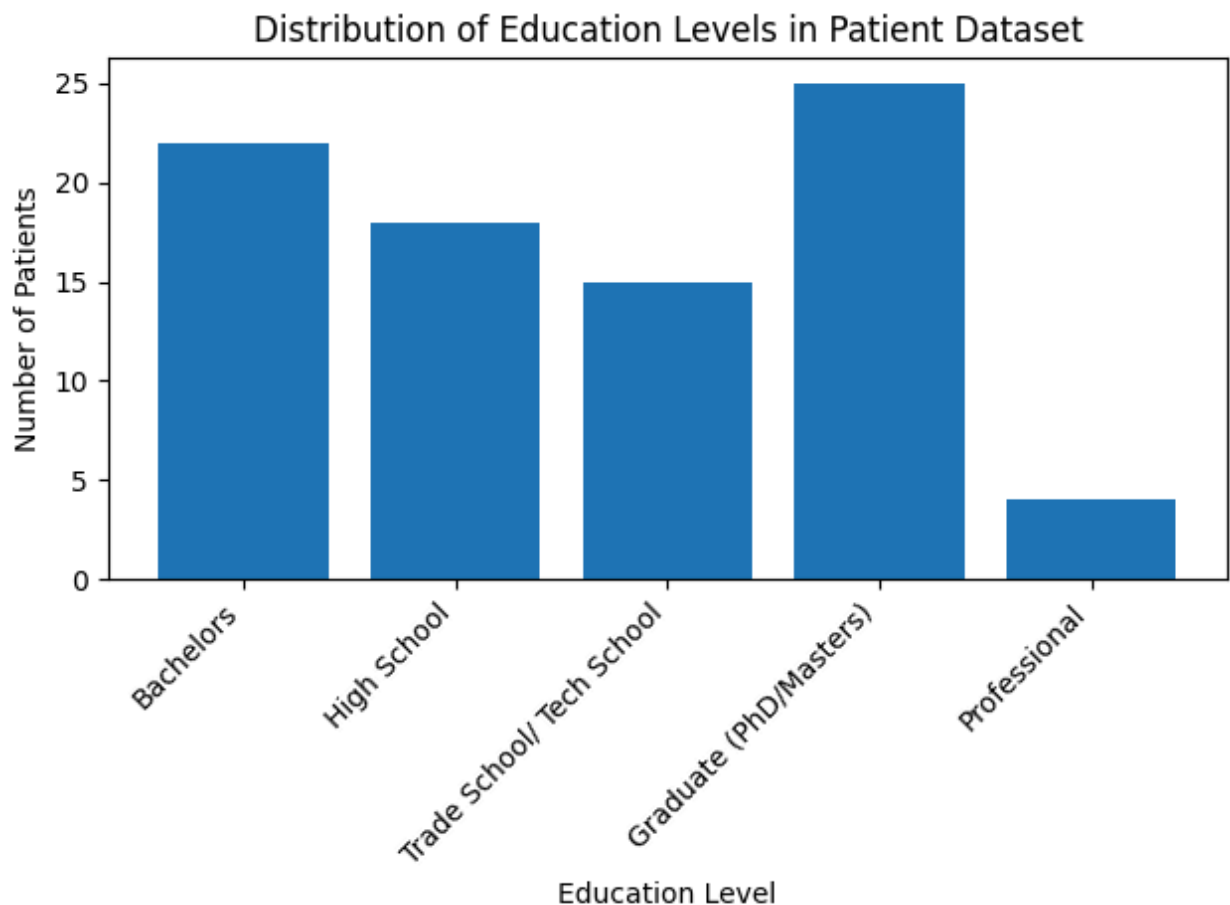
# Load patients
Patient.instantiate_from_csv('UpdatedLuminex.csv', 'UpdatedMetaData.csv')

# Build dictionary grouped by education
Patient.sort_ed()
Patient.subsort_thal()

# Count patients in each education level
education_counts = {lvl: len(pats) for lvl, pats in Patient.education_lvl.items()}

# Plot a bar chart of education levels
plt.bar(education_counts.keys(), education_counts.values())
plt.xticks(rotation=45, ha="right")
plt.xlabel("Education Level")
plt.ylabel("Number of Patients")
plt.title("Distribution of Education Levels in Patient Dataset")
plt.tight_layout()
plt.show()

```



```
In [33]: import matplotlib.pyplot as plt
import numpy as np
from scipy import stats

# Prepare labels
labels = list(Patient.education_lvl.keys())

# Compute mean and standard deviation of Thal scores for each education level
mean_thal_scores = []
std_devs = []

for lvl in labels:
    thal_scores = [p.thal_score for p in Patient.education_lvl[lvl] if p.thal_
if thal_scores:
    mean_thal_scores.append(np.mean(thal_scores))
    std_devs.append(np.std(thal_scores))
else:
    mean_thal_scores.append(0)
    std_devs.append(0)

mean_thal_scores = np.array(mean_thal_scores)
std_devs = np.array(std_devs)

# Error bars: only upward
lower_errors = np.zeros_like(std_devs)
```

```

upper_errors = std_devs

# Plot mean Thal scores with error bars
plt.bar(labels, mean_thal_scores, yerr=[lower_errors, upper_errors], capsize=5)
plt.xticks(rotation=45, ha="right")
plt.xlabel("Education Level")
plt.ylabel("Mean Thal Score")
plt.title("Mean Thal Scores by Education Level with Standard Deviation")
plt.tight_layout()
plt.show()

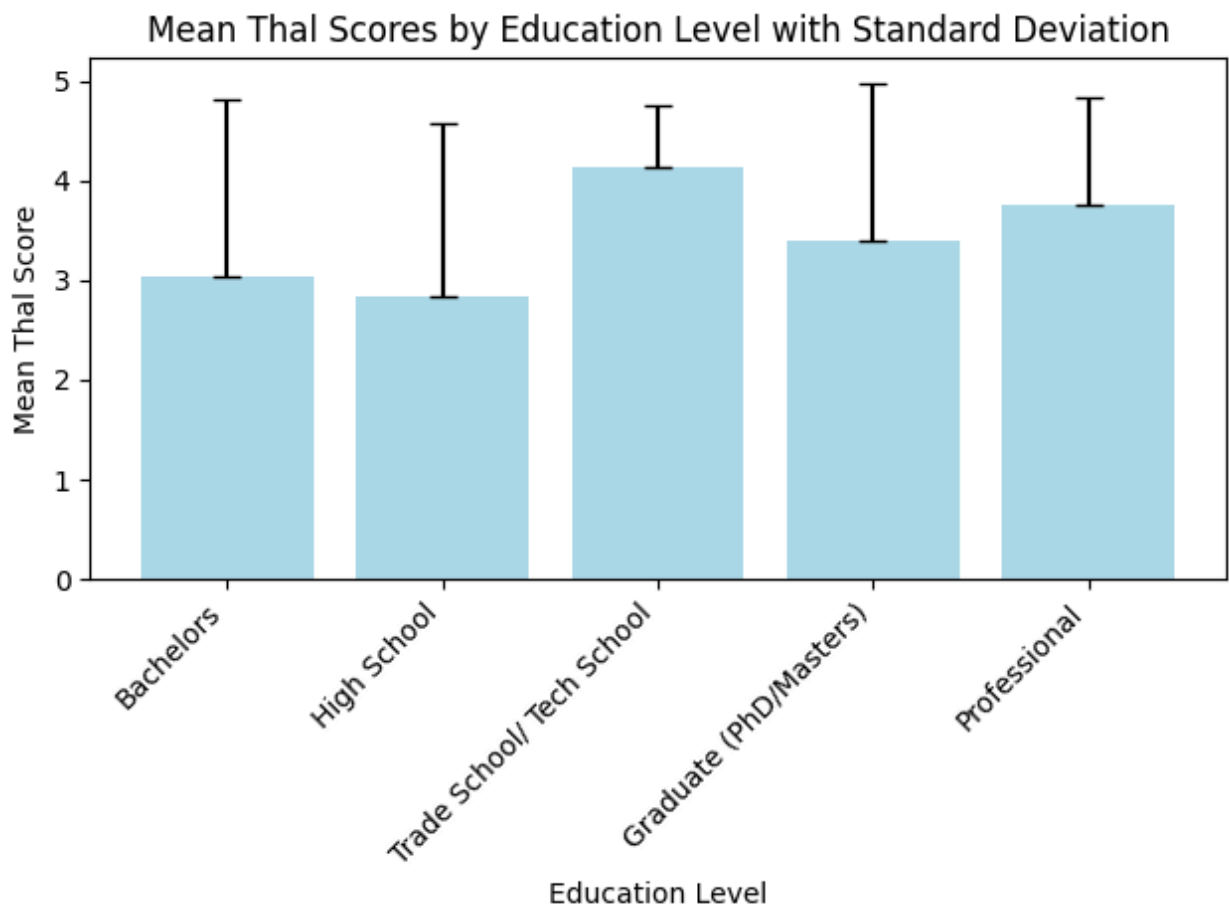
# =====
# ANOVA: Compare Thal scores between education levels
# =====

# Prepare groups of Thal scores by education level
thal_groups = []
for lvl in labels:
    thal_scores = [p.thal_score for p in Patient.education_lvl[lvl] if p.thal_
    if thal_scores:
        thal_groups.append(thal_scores)

# Perform one-way ANOVA
anova_result = stats.f_oneway(*thal_groups)

# Print results
print("ANOVA Results:")
print(f"F-statistic: {anova_result.statistic:.3f}")
print(f"P-value: {anova_result.pvalue:.3e}")
if anova_result.pvalue < 0.05:
    print("Result: Significant differences in Thal scores between education le
else:
    print("Result: No significant differences in Thal scores between education

```



ANOVA Results:

F-statistic: 1.686

P-value: 1.615e-01

Result: No significant differences in Thal scores between education levels (fail to reject null).

In [34]: `import matplotlib.pyplot as plt`

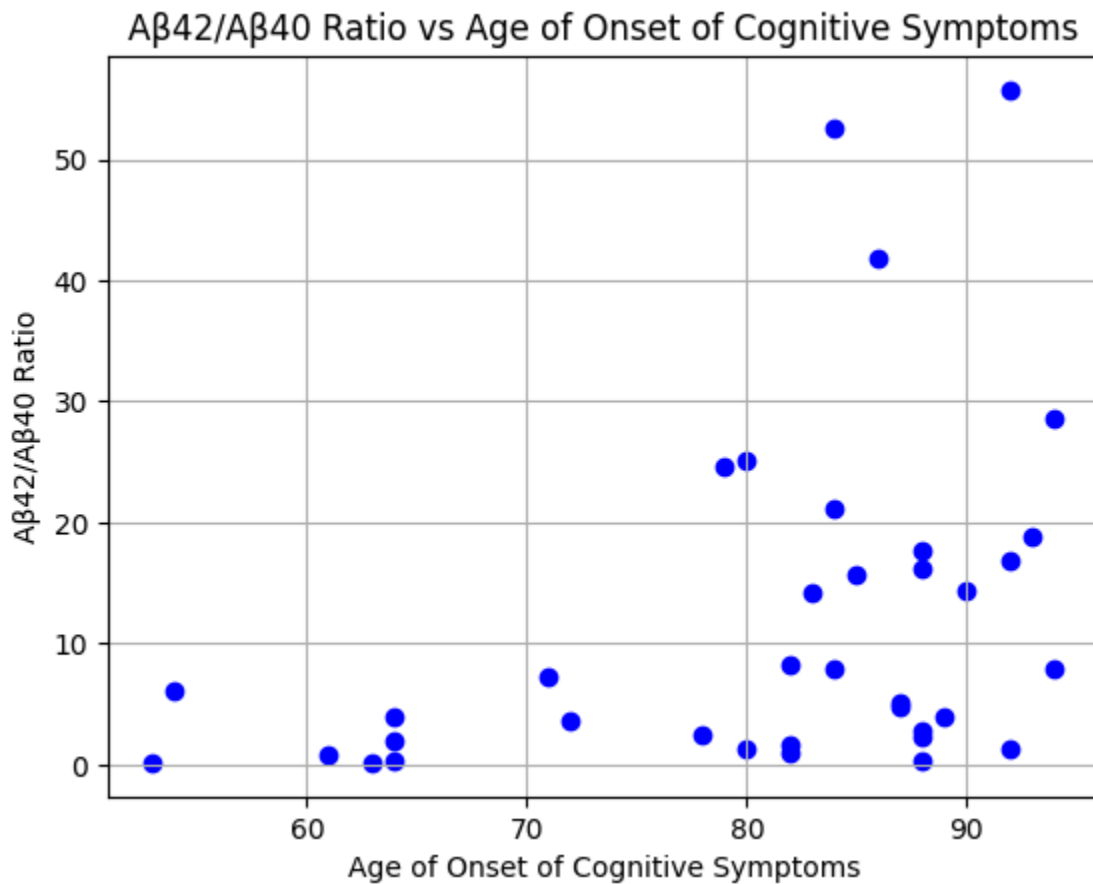
```
# Initialize lists
onset_age_list = []
ab_ratio = []

# Process patient data
for patient in Patient.all_patients:
    onset_age_list.append(patient.age_symp_on)

    if patient.ABeta40:
        ratio = patient.ABeta42 / patient.ABeta40
        ab_ratio.append(ratio if ratio <= 500 else None) # Filter out unrealistic
        '''
        one data point had a AB40/AB42ratio of around 2000, which is is around
        decided to filter it out for the sake of
        '''
    else:
        ab_ratio.append(None) # Use None for missing or invalid data
```

```
# Filter out None values for plotting
filtered_data = [(age, ratio) for age, ratio in zip(onset_age_list, ab_ratio)
ages, ratios = zip(*filtered_data)

# Scatter plot
plt.scatter(ages, ratios, color='blue')
plt.xlabel("Age of Onset of Cognitive Symptoms")
plt.ylabel("Aβ42/Aβ40 Ratio")
plt.title("Aβ42/Aβ40 Ratio vs Age of Onset of Cognitive Symptoms")
plt.grid(True)
plt.show()
```



```
In [35]: # Saving data of ABeta42/ABeta40 ratio and Age of Onset to CSV
import pandas as pd
print(ab_ratio)
print(onset_age_list)
# Create a DataFrame
df = pd.DataFrame({
    'Age of Onset': onset_age_list,
    'ABeta42/ABeta40 Ratio': ab_ratio
})
# Write to CSV
df.to_csv('patient_data.csv', index=False)
print("CSV file 'patient_data.csv' has been created.")
```

[49.517166433422304, 12.71707315065794, 246.12665267312826, 1.3209014689550918, 3.145286886329164, 25.51133966084625, 5.108504398141206, 18.861423220879022, 2 1.981325863304452, 30.67060619343318, 14.200866269767332, 7.96296296386408, 3.9 78464306577392, 7.180158596466259, 16.76240049369563, 0.3148153516877255, 20.62 5697163563785, 27.11650774721221, 2.0077366856966194, 16.25242718014671, 16.806 722688926552, 17.662371127610783, 3.3834352081647405, 21.21218487702923, 0.4823 717030059996, 6.103848759426092, 1.034467618024906, 17.423184604374086, 28.5016 10549882916, 1.2762003096923038, 15.673241156347762, 13.255742725874612, 8.9888 22654657646, 43.930041939876, 0.623248304805726, 34.60053487229384, 0.145477480 12112766, 2.7668012153961055, 458.9892235887318, 0.7982425584136721, None, Non e, 3.9586776851318475, 6.443554317992643, 25.169745571740908, 1.0, 0.2123663986 5584228, 0.3689644845167863, 199.51071800540657, 16.229268268201068, 1.58132060 77447084, 7.252109932628356, 15.27229297832975, 55.720569906638964, 14.28688524 3133872, 41.755999948244, 18.205434800762323, 24.660544219546278, None, None, 1 91.9690706125645, 0.4168941387422491, 3.4262097655014423, 1.2562570537030924, 2.8188579356437353, 8.238013701183162, 10.149769584969752, 3.61419224576424, 4.926549348639659, 8.006102413345795, 13.254806856628043, 19.150063226317624, 1 4.294222751269434, 12.415470294179709, 1.539073308821431, 52.55079825108116, 7.524357338928067, 17.596848737916098, 2.3337433373052168, 18.75693526589127, 4.717449070277016, 2.5149862427681526, 7.928695649273271, 49.9999816523674] [None, None, None, 80, None, None, 87, None, None, None, 83, 84, 64, 71, None, 64, None, None, 64, None, 92, 88, None, 84, None, 54, None, None, 94, 92, 85, N one, None, None, None, None, 63, 88, None, 61, None, None, 89, None, 80, 82, 5 3, 88, None, 88, 82, None, None, 92, 90, 86, None, 79, 93, None, None, None, No ne, None, None, 82, None, 72, None, None, None, None, None, None, None, 84, Non e, None, 88, 93, 87, 78, 94, None]

CSV file 'patient_data.csv' has been created.

```
In [36]: import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
import pandas as pd

# Load and clean data
df = pd.read_csv("patient_data.csv")
df_clean = df.dropna(subset=["Age of Onset", "ABeta42/Abeta40 Ratio"])
x = df_clean["Age of Onset"].values.reshape(-1, 1)
y = df_clean["ABeta42/Abeta40 Ratio"].values

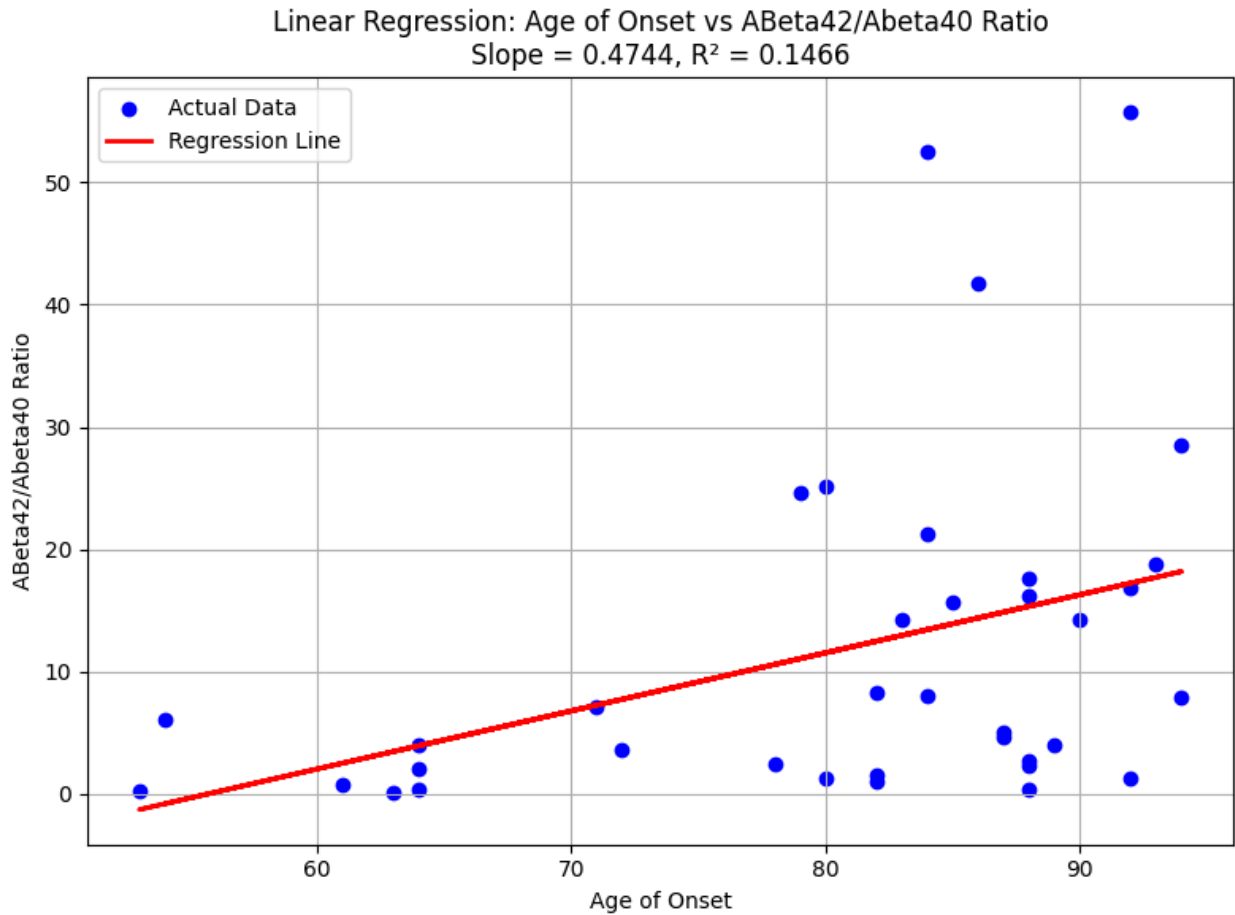
# Fit model
model = LinearRegression()
model.fit(x, y)

# Predict values for regression line
y_pred = model.predict(x)

# Get slope and R2
slope = model.coef_[0]
intercept = model.intercept_
r_squared = model.score(x, y)

# Plotting
plt.figure(figsize=(8, 6))
plt.scatter(x, y, color='blue', label='Actual Data')
plt.plot(x, y_pred, color='red', linewidth=2, label='Regression Line')
```

```
plt.xlabel("Age of Onset")
plt.ylabel("ABeta42/Abeta40 Ratio")
plt.title(f"Linear Regression: Age of Onset vs ABeta42/Abeta40 Ratio\nSlope = ")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```



Verify and validate your analysis:

The graphical analyses did not show a clear correlation between education level and AD diagnosis or severity. This observation was supported by the statistical tests we conducted in which the calculated R and p values indicated no significant association between years or level of education and outcome of the disease.

Each statistical test was selected to ensure the analysis of the data was proper.

- Correlation tests were used to measure the strength and direction of a potential linear relationship between education and AD diagnosis.
- ANOVA was applied for a comparison of means across different education groups to determine and significant differences in AD

prevalence.

All tests consistently showed that there was no statistically significant difference between groups.

Other publications:

- <https://academic.oup.com/aje/article/159/2/175/166359>
- https://www.neurology.org/doi/pdfdirect/10.1212/WNL.43.1_Part_1.13
- <https://jamanetwork.com/journals/jama/article-abstract/368856>

Large epidemiological studies show that education is not a consistent independent predictor in AD progression or severity. The JAMA cohort study found lower education linked with higher AS risk, but this effect overlapped with occupation and socioeconomic status which suggests that education is not a factor on its own. Karp et al. (2004) similarly observed that education remained significant only when adult socioeconomic status was excluded. A systematic review across 88 study populations reported that 42% found no significant relationship between education and the progression of dementia. All together, these findings validate what we concluded based on our dataset analysis: the lack of correlation between education and AD severity reflects evidence that education alone is insufficient to reliably predict risk. Outcomes are shaped by additional lifestyle and biological factors.

Conclusions and Ethical Implications:

Conclusions:

- The analysis of the dataset revealed no prominent or consistent relationship between a patient's level of education and the progression/outcome of AD. The examined dataset does not reflect our original hypothesis that education may contribute to cognitive reserve.

Ethical Implications:

- An ethical implication regarding the interpretation of the data is that there is a sense of responsibility to not overstate weak or inconclusive findings. The suggestion of correlation between a related neurological factor (such as education level) could mislead patients and their families, or even policymakers.
- It is important for providers to approach patients holistically, ensuring that one factor, detail, or number does not fully drive their decision making process.

Limitations and Future Work:

- Education alone is very unlikely to explain Alzheimer's risk or progression. Future work should also incorporate any genetic biomarkers, sociological/lifestyle factors, and other common measures when dealing with AD.
- A longitudinal dataset in particular would be very valuable since AD gradually develops over what could be a period of many years.
- Future work in this area could also explore interactive effects in the sense of pairing the analysis of education level with diet, access to healthcare, access to further education, or even physical activity. This exploration could provide a more thorough conclusion while simultaneously relating education level to AD severity like the original project intended.

NOTES FROM YOUR TEAM:

Began reviewing the SEA-AD dataset and set up Python environment. Wrote initial code for loading and inspecting the dataset (checked donor demographics, age distribution, etc.). Discussed project focus: relationship between Alzheimer's disease progression and education level.

9/19: We processed and structured the dataset, visualizes patient distributions via a bar graph, and completed an ANOVA to assess the influence of education on Thal scores. All of this progress was based on the determined project focus of the relationship between AD progression and education level. The ANOVA provided a foundation for a deeper investigation into our project focus.

QUESTIONS FOR YOUR TA:

check-in 1: none at the moment, but struggling a bit with GitHub

check-in 2: no new questions; still have a few concerns about GitHub/VSCode. A few struggles every now and then but nothing unmanageable.

check-in 3: no questions!